Aaradhya Dev Tamrakar

AI Safety & Fairness Auditing · Model Evaluation · Applied Machine Learning

Kathmandu, Nepal • aaradhyadevtmr@gmail.com • +977 9844602050 • linkedin.com/in/aaradhya-dev-tamrakar • github.com/AaradhyaDT • aaradhyadt.github.io • scholar.google.com

Final-year Electronics & Information Engineering undergraduate specializing in applied AI fairness auditing and empirical model evaluation. Fuse AI Fellow co-developing diagnostic prediction-auditing frameworks measuring canonical demographic parity and error-rate disparities across intersectional cohorts (AIF360, Fairlearn). Experienced in statistical evaluation, edge classifier validation on wearable sensor data, and reproducible benchmarking.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Co-developed **BiasAperture**, an automated demographic fairness auditing pipeline measuring predictive disparities on computer vision benchmarks as the fellowship capstone project.
- Developed customer churn and lifetime value (CLV) prediction workflows utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026

Python · AIF360 · Fairlearn · Statistical Testing · Jinja2 · OpenCV

- Co-developed an automated prediction-auditing and reporting pipeline as a two-person fellowship capstone with Tisha Manandhar.
- Audited classification performance across **126 demographic bins** on the FairFace benchmark (**97,698 images**), computing Disparate Impact Ratio, Demographic Parity Difference, and Equalized Odds.
- Engineered self-contained HTML and PDF report generation with confidence intervals and diagnostic disparity plots; matched model comparison and validation remain active work.

SPARK — Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TensorFlow Lite Micro · Python · FreeRTOS · MPU-6050

- Trained 1D-CNN fall-detection models on the SisFall benchmark (38 subjects, 38,420 windows), evaluating subject-stratified cross-validation with **0.9185 AUC-ROC** and 87.81% sensitivity.
- Quantized model to an **18.5 KB INT8 CNN** footprint for ESP32-S3 edge deployment; on-device hardware testing, BLE streaming, and volunteer cohort validation remain pending.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

TECHNICAL SKILLS

Machine Learning & Evaluation: Python, PyTorch, scikit-learn, Pandas, NumPy, XGBoost, SHAP, Fairlearn, AIF360, OpenCV, MediaPipe

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX

CERTIFICATIONS, WORKSHOPS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) • IEEE SPAx (Student Professional Awareness Venture) Organizing Committee (2025 – 2026) • IEEEXtreme 19.0 Programming Competition (2025) • PCB Design & Fabrication Workshop (2025) • AWS Cloud Computing Workshop (2024) • Prompt Engineering Fundamentals (2024) • Microsoft Learn Student Ambassador (2024 – 2025)

ACADEMIC & PROFESSIONAL REFERENCES

Er. Suramya Sharma Dahal — Head of Department, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University) • Email: surammya@gmail.com. Context: Academic department head overseeing undergraduate engineering curriculum and senior projects

Er. Dipen Manandhar — Faculty Member & Major Project Supervisor, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University). Context: Undergraduate major capstone project supervisor (SPARK edge-AI wearable)

Shreejan Kisee — Teaching Assistant & Capstone Supervisor

Fusemachines AI Fellowship. Context: AI Fellowship capstone supervisor (BiasAperture demographic auditing) and coursework mentor